NeuraEdge Universal
A Beginner-to-RTL Explanation of the Neuromorphic Transformer Accelerator

Generated from the implemented RTL, Python runtime, tests, and Vivado flow

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1 What the machine is

NeuraEdge Universal is a programmable arithmetic accelerator for the `xc7z020clg400-1` FPGA on a PYNQ-Z2 board. It is not wired to one image size or one classifier. Software describes work using 128-bit commands, and the same physical engines can execute matrix multiplication, convolution lowered to matrix multiplication, binary and ternary spiking arithmetic, residual/vector operations, integer normalization, LIF neuron updates, lookup-table functions, and fixed-point SGD with momentum.

For the reproduced CIFAR-10 workload, the source four-timestep Spikingformer checkpoint classifies 9,555 of 10,000 official test images correctly (95.55%). The exported integer-coefficient arithmetic classifies 9,542 images correctly (95.42%). The difference is 13 images, or 0.13 percentage points. These numbers are complete software test-set evaluations. RTL operator tests, archive-integrity tests, command tests, and routed Vivado reports form separate evidence and are described later rather than being mixed into the accuracy number.

2 Electronics and FPGA concepts from zero

2.1 A bit is a voltage range

Digital logic calls a low voltage "0" and a high voltage "1". A wire in SystemVerilog describes a physical connection that will carry one of those levels after synthesis. A bus such as `logic [63:0] data` is 64 parallel wires. It can represent an unsigned integer, a signed two's-complement integer, eight bytes, or control fields; the surrounding logic determines the meaning.

2.2 Combinational logic and registers

Combinational logic is made from lookup tables, carry chains, and dedicated arithmetic blocks. Its output changes after its input changes. A register is a bank of flip-flops that copies its input only at a clock edge and remembers the copied value between edges. The accelerator period is 5 ns, so its target frequency is

$$f = \frac{1}{5 \times 10^{-9}} = 200 \text{ MHz}.$$

The complete board design also has an 8 ns (125 MHz) DMA/DDR clock and a 10 ns (100 MHz) control clock. Separate clocks are like workers stepping to different metronomes: a multibit value cannot simply be sampled between them. AMD AXI clock converters therefore use handshake synchronizers and asynchronous FIFOs to transfer complete transactions without mixing bits from different cycles.

An `always_comb` block describes combinational gates. An `always_ff @(posedge clk)` block describes registers updated at the rising edge. The nonblocking operator `<=` means all right-hand sides are sampled from the old cycle and all left-hand registers update together. This is why the order of nearby nonblocking assignments does not turn them into sequential software statements.

2.3 Finite-state machines

A finite-state machine stores a small state code in registers. Combinational decode logic selects what happens in each named state. For example, the descriptor sequencer moves through `HEADER`, `COMMAND_LOW`, `COMMAND_HIGH`, `SUBMIT`, `PAYLOAD`, and `WAIT`. Each transition consumes at least one clock edge, so the waveform exposes the exact protocol progress.

2.4 Valid and ready

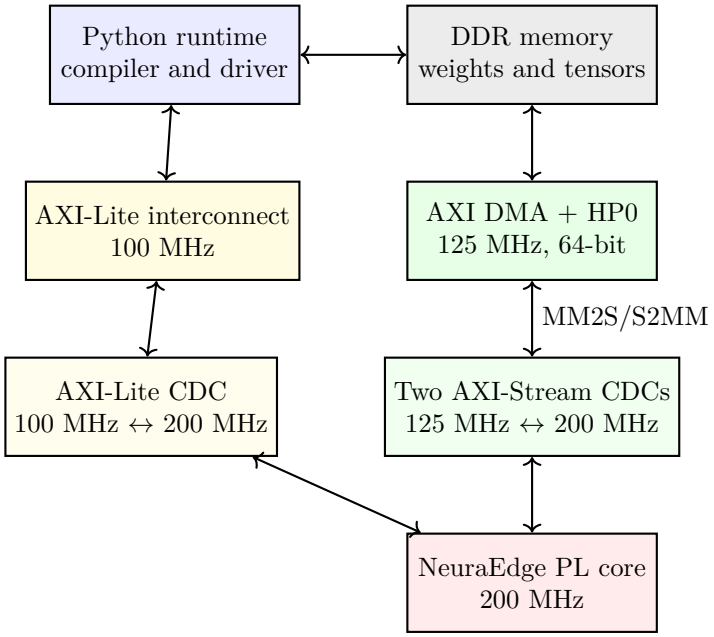
AXI-Stream uses a two-sided handshake. The producer raises `TVALID` when its data is meaningful. The consumer raises `TREADY` when it has space. A transfer occurs only on a rising clock edge where both are 1:

$$\text{transfer} = \text{TVALID} \wedge \text{TREADY}.$$

If TVALID=1 and TREADY=0, the producer must hold data and TLAST stable. This is backpressure. The testbenches deliberately stall ready to prove results are neither lost nor duplicated.

3 Board-level system

The Zynq-7020 contains a processing system (PS) and programmable logic (PL) on one chip. The PS is the dual-core ARM processor running PYNQ/Linux and Python. The PL is the FPGA fabric containing NeuraEdge. External DDR is connected to the PS and stores the complete model and large tensors. AXI DMA moves contiguous buffers between DDR and the accelerator without the ARM copying each word.



There are three reset controllers, one per domain. Reset assertion brings each island to a known state; deassertion is synchronized to that island's own clock. The accelerator never samples a raw 100 or 125 MHz bus. The two stream clock converters carry TDATA, TKEEP, TLAST, and their valid/ready state as coherent transactions.

The full Vivado block design uses a 26-bit AXI DMA length field. One legal DMA packet can therefore contain at most

$$2^{26} - 1 = 67{,}108{,}863 \text{ bytes.}$$

The software driver checks that limit. A larger descriptor program is split only between records, never through a command header or payload. The PL retains its model/neuron state between those packets.

3.1 What Vivado physically implemented

Vivado 2026.1 routed the complete block design, generated the BIT/HWH/XSA, and reported zero routing errors. This is the final board result, not the earlier out-of-context estimate:

Measured routed quantity	Value
Worst setup slack / total setup violation	+0.005 ns / 0.000 ns
Worst hold slack / total hold violation	+0.050 ns / 0.000 ns
Used LUTs / flip-flops	45,245 / 51,077
DSP48E1 / RAMB36 / RAMB18	110 / 33 / 8
Fully routed nets / routing errors	74,557 / 0
Vectorless total / dynamic estimate	2.152 W / 1.993 W
SAIF total / dynamic estimate	1.937 W / 1.785 W
SAIF full-design net match / confidence	9,164 of 123,250 / Medium

The SAIF estimate attributes 0.472 W dynamic power to the accelerator hierarchy. Untouched memories produce explicit X-state warnings, so the 93% unmatched portion is filled probabilistically by Vivado. These values are useful implemented-design estimates; they are not relabeled as readings from a physical board supply rail.

4 Descriptor programs

Issuing one AXI-Lite transaction per tile would make the ARM part of every inner loop. Descriptor mode instead sends a complete program through DMA. Each record contains the following 64-bit beats:

`Beat`	Exact contents and purpose
`header`	Bits 31:0 are payload beat count; bits 62:32 are result beat count; bit 63 marks the final record of this DMA program.
`command low`	Command bits 63:0.
`command high`	Command bits 127:64. If the command has no payload and this is the final record, input TLAST is asserted here.
`payload`	Exactly the declared number of 64-bit input beats. The sequencer regenerates the per-command TLAST expected by the selected core engine.

The sequencer has a one-beat elastic output register. It can pop the old beat and capture a new core beat in the same cycle, sustaining one result per cycle under no backpressure. Declared output counters are deliberately kept out of the core-ready path. An extra core result is accepted, discarded locally, and reported as a program error; it cannot deadlock the core or leak into DDR.

5 Command-driven universal core

`neuraedge_universal_core.sv` decodes the opcode and connects the shared input/output streams to exactly one engine. The important engine classes are:

Engine	Physical responsibility
Buffered tensor core	Stores activation/weight tiles, compacts sparse K positions, schedules dense/binary/ternary work, and streams 32-bit results.
Vector ALU	Residual add, maximum, comparisons, ReLU, spike conversion, signed rounding, saturation, and accumulator requantization on eight lanes.
Alpha-XNOR	Spiking self-attention score using matching-one and matching-zero events.
LayerNorm	Integer mean, variance, integer square root, division, learned gamma/beta, and signed output saturation.
Log softmax	Integer power-of-two probability approximation and normalization.
LIF bank	32 persistent membrane states with leak, threshold, reset, and 64-bit spike telemetry.
Activation LUT	Eight-lane programmable lookup table for nonlinear functions.

Engine	Physical responsibility
SGD optimizer	Eight-lane signed parameter update with momentum, saturation, and update telemetry.

Only the selected engine sees command-valid. Input ready comes from that engine, and the output arbiter returns only that engine's valid/data/last. A busy engine must complete before another command is accepted. This simple ownership rule prevents two engines from driving the same stream simultaneously.

6 Dense tensor arithmetic

6.1 Matrix multiplication

For matrices $A \in \mathbb{Z}^{M\times K}$ and $B \in \mathbb{Z}^{K\times N}$, the output is

$$C_{m,n} = \sum_{k=0}^{K-1} A_{m,k}B_{k,n}.$$

The physical tile computes eight output rows and sixteen output columns. It has 64 DSP48E1 blocks, each producing two exact byte products, so 128 byte MAC lanes are available per cycle. The arithmetic peak at 200 MHz is

$$128 \times 200 \times 10^6 = 25.6 \text{ GMAC/s}.$$

Counting multiply and add as separate operations gives 51.2 GOPS. This is a peak issue rate, not an application throughput claim: loading, boundaries, sparsity, vector operations, and DDR can reduce sustained throughput.

6.2 Two dataflows

Outer mode broadcasts one K-position activation per row and reuses sixteen weights across the columns. It is useful for ordinary matrix tiles with several rows. Dot8 mode feeds eight adjacent K positions into each column and reduces them, keeping the same DSPs useful when the output has only one row. The compiler estimates byte traffic for legal schedules and chooses a mode rather than fixing one traversal for every layer.

6.3 Exact asymmetric quantization

An unsigned byte A represents real zero at activation zero point Z_a; an unsigned byte B represents real zero at weight zero point Z_b. Direct centered arithmetic is

$$\sum_k (A_k - Z_a)(B_k - Z_b) + \text{bias}.$$

Expanding the product lets hardware use unsigned DSP products plus row and column corrections:

$$\underbrace{\sum_k A_k B_k}_{\text{raw}} - Z_b \underbrace{\sum_k A_k}_{R_a} - Z_a \underbrace{\sum_k B_k}_{C_b} + K Z_a Z_b + \text{bias}.$$

The Python golden test evaluates both equations on randomized shapes and zero points. Equality is exact integer equality; no tolerance hides a mismatch.

6.4 Two byte products in one DSP

The packed weight is

$$P = B_{lo} + (B_{hi} \ll 17).$$

Multiplying P by activation A places AB_{lo} in bits 15:0 and AB_{hi} in bits 32:17. Bit 16 is a guard because $255 \times 255 = 65025 < 2^{16}$. The high packed bit can make the DSP operand appear signed, so the RTL adds an exact correction through the DSP C input. Vivado's synthesis log confirms the intended $C + A \times B$ DSP mode, and the testbench exhaustively checks 131,072 packed input combinations.

7 Event-driven arithmetic

Binary weighted mode implements

$$y_n = \sum_k s_k w_{k,n}, \qquad s_k \in \{0, 1\}.$$

When $s_k = 0$, no weight is added. Ternary mode accepts $t_k \in \{-1, 0, +1\}$ and conditionally subtracts, skips, or adds the weight. The activation loader detects K positions that are zero for every active row and stores only their surviving indices. Compute then iterates over the compact list, so an all-zero reduction position removes a cycle instead of merely presenting zero to a multiplier.

Two hardware counters separate ordinary dense MACs from event-driven weight adds. They are per-command exact counts, accumulated into wider wrapper telemetry. This makes activity visible in waveforms and software snapshots.

8 LIF neurons

A leaky-integrate-and-fire neuron retains membrane $u[t]$ between timesteps. The implemented fixed-point order is:

$$\begin{aligned} u_{leak}[t] &= \text{roundShift}(u[t-1]\,\lambda), \\ u_{int}[t] &= u_{leak}[t] + I[t], \\ s[t] &= [u_{int}[t] \geq \theta], \\ u[t] &= \begin{cases} u_{reset}, & s[t] = 1 \\ u_{int}[t], & s[t] = 0. \end{cases} \end{aligned}$$

$I[t]$ is input current, λ is leak, θ is threshold, and square brackets produce 1 when the condition is true. The bank stores 32 membranes. Software chunks a larger logical tensor into banks and executes all timesteps for one chunk before reusing the physical lanes. Clearing membrane state is separate from clearing lifetime spike counters, so image boundaries do not erase telemetry accidentally.

9 Spiking attention and transformer operations

Alpha-XNOR compares binary query and key spikes. A matching one contributes 256 in Q8 units; a matching zero contributes programmable α_{Q8}:

$$score = 256 N_{11} + \alpha_{Q8} N_{00}.$$

The ordinary binary QK path used by the exported checkpoint is also supported: conditional additions form the attention score, a threshold LIF produces an attention event, unsigned AV aggregation accumulates values, and a vector requantization applies the exact 0.125 scale. Dense QK, integer LayerNorm, power-of-two softmax, residual add, and MLP projections remain available for standard/hybrid transformer graphs.

10 Exact exported Spikingformer graph

The executable runtime follows this checkpoint's actual forward function:

1. Normalize one $3 \times 32 \times 32$ CIFAR-10 image.
2. Execute five folded tokenizer convolutions. The first projection is computed once and broadcast across four timesteps, matching source code.
3. Apply LIF after each tokenizer stage and binary max-pool at the two spatial reduction points, ending at 384 channels and 4×4 positions.
4. For each of four transformer blocks, compute Q, K, and V, reshape into 12 heads, run spiking QK and AV, apply the output projection, and add the attention residual.
5. Execute the $384 \rightarrow 1536 \rightarrow 384$ spiking MLP and add its residual.
6. Average over four timesteps and 16 spatial positions, then execute the signed INT8 $384 \rightarrow 10$ classifier and argmax.

The source checkpoint's forward function does not call LayerNorm, so the runtime does not insert one. Operator support and model use are different facts: the accelerator supports LayerNorm for other transformer graphs, while this exact model does not request it.

The generated manifest totals 3,732,164,352 model MAC equivalents, 104,256 tile commands, 243,330,192 raw tile-input bytes, and 52,297,808 result values. These counts come from real exported shapes and the compiler schedule.

11 Parameter archive and corruption checks

The hardware archive is 9,385,704 bytes and has SHA-256 digest `b02c5df519eaecffb228f4c8be21babd d0c01761 f26210bb8dfcca3318ed0d381`. It contains 118 64-byte-aligned sections representing 29 folded projections and the classifier. The loader verifies the whole-file size and digest, section order and uniqueness, non-overlap and bounds, dtype/shape byte counts, each section digest, and every projection reference. Returned NumPy views are read-only so execution cannot silently modify trained parameters.

12 Host compiler and memory behavior

The compiler validates dimensions before allocating DMA buffers. Convolution is lowered to rows of receptive-field values and then tiled as GEMM. Group, stride, padding, dilation, depthwise binary, and depthwise ternary cases have directed tests. The compact stream representation stores each 64-bit beat once in a standard-library integer array instead of expanding millions of Python objects.

For the real block-3 shape $1024 \times 1728 \times 384$, the recorded benchmark produces 6,168 instructions, 43,130,880 payload bytes, 148,032 descriptor bytes, 43,278,912 total input bytes, and 3,145,728 output bytes. Vectorized packing took 0.334 s in the recorded run; the former scalar implementation took 95.601 s for the same case.

13 Full-backbone on-device training flow

The training entry point uses the same exported 29 projections and classifier as inference. A minibatch follows this exact order:

1. Read a hashed official CIFAR-10 shard, apply a deterministic reflected crop and horizontal flip, and normalize the resulting RGB samples.
2. Run the complete quantized forward graph while retaining projection inputs, pre-reset LIF membranes, spikes, pool inputs, attention operands, and residual branch values needed by the reverse pass.

3. Evaluate the ten-way power-of-two softmax in programmable logic. The vector ALU forms the Q15 batch-mean classifier derivative $p - \text{onehot}(y)$.

4. Traverse the graph in reverse: classifier, four transformer blocks, tokenizer, and finally the input image. Every converging residual branch is added explicitly and every parameter name must appear exactly once.

5. Convert each block-floating parameter gradient to Q15, clamp it in the vector fabric, update its signed master and momentum tensors in the optimizer lanes, and redeploy the rounded/clamped inference tensor.

6. Atomically checkpoint every master tensor, every momentum tensor, optimizer step, epoch, next minibatch index, graph identity, and numeric format before a pause or according to the configured checkpoint interval.

The complete 9.3 MB parameter state cannot fit in the Zynq-7020's roughly 630 KiB of block RAM. Master tensors and saved activations therefore reside in board DDR, while matrix backward, vector gradient algebra, nonlinear derivatives, and parameter updates execute in programmable logic. DMA moves bounded tiles; it does not replace an FPGA arithmetic result with a NumPy calculation.

The generated coverage manifest derives its counts from the exported graph. It contains 43 of 43 execution operators, 30 learned affine groups, eight dynamic attention matrix products, 32 LIF sites, two max-pool sites, and eight residual fan-outs, with zero uncovered operators. One full sample schedules 14,742,724 backward partial tiles and 7,474,847,252 signed backward multiplications across 9,295,978 deployed trainable parameter elements.

13.1 Exact matrix and convolution derivatives

For one affine layer

$$Y = XW + b,$$

the reverse equations implemented by the training compiler and signed backward array are

$$dX = dYW^T, \qquad dW = X^T dY, \qquad db = \sum_m dY_m.$$

The compiler appends a column of ones to X and a bias row to W, so the same verified matrix engine produces dW and db without a host-side reduction. Arbitrary dimensions are partitioned into at-most 8×8 external tiles. For dX, one row/input tile remains in the 47-bit accumulator while every output-feature tile is traversed. For dW, one input/output tile remains while every sample-row tile is traversed. Intermediate commands suppress output; only the last reduction chunk performs one signed round and streams the final tile.

Inside `neuraedge_linear_backward8.sv`, a physical 4×4 signed DSP array visits two result-column groups and, when needed, two reduction groups. Sixteen registered 16×16 products become two 33-bit pair sums, one 34-bit four-product sum, and a mathematically sufficient 35-bit local sum. That sum is sign-extended only when added to the retained 47-bit accumulator. Magnitude, rounding bias, shift, sign restoration, saturation, and result storage occupy separate states so no combinational carry chain spans all those operations at 200 MHz.

Convolution backward uses the same equations after exact signed `im2col`. Group, stride, padding, dilation, and kernel coordinates are retained in the compiled plan. The returned patch gradient is scattered back with saturating overlap addition; grouped and depthwise shapes never mix channels from different groups.

13.2 LIF, pooling, attention, and residual derivatives

Hard thresholding has zero derivative almost everywhere, so training uses the SpikingJelly sigmoid surrogate with $\alpha = 4$. A generated 256-entry Q15 table stores

$$\sigma'(x) = \alpha\sigma(\alpha x)(1 - \sigma(\alpha x)).$$

The address is the saturated, rounded pre-reset membrane distance from the threshold. Reverse-time BPTT walks the four cached timesteps from 3 to 0. With detached hard reset, the temporal membrane path is cleared where the

forward step spiked; leak, surrogate multiplication, and input scaling are Q15 vector operations with explicit signed saturation.

Max-pool backward reproduces PyTorch's first-maximum tie rule, then uses fabric maximum/comparison and scatter-add operations to return gradients to the chosen input coordinates. Dynamic QK and AV products use the same two matrix-backward passes as learned projections. Attention's exact factor 0.125 is a power-of-two gradient shift. Residual addition copies the incoming derivative to both parents; derivatives from branches that meet later are combined by the saturating vector adder.

14 On-chip parameter update engine

The SGD engine accepts current signed weights w, signed gradients g, and momentum state v. Its fixed-point form is

$$\begin{aligned} v' &= \text{roundShift}(\mu v) + g, \\ d &= \text{sat}\left(v' + \text{roundShift}(\lambda w)\right), \\ w' &= \text{sat}\left(w - \text{roundShift}(\eta d)\right). \end{aligned}$$

μ is momentum, η is learning rate, and λ is weight decay. The retained velocity v' contains the loss gradient and momentum only; decay enters the temporary direction d and therefore cannot accumulate in the momentum state. This is the SGDW decoupling defined by Loshchilov and Hutter. Rounding is symmetric for positive and negative values, and saturation clamps rather than wrapping at the signed limits. The update loop processes eight lanes, writes both new weight and velocity state, and increments an exact update counter. The same tensor engine can execute matrix products needed for gradient formation; the optimizer is the fabric-resident primitive that applies those gradients.

15 Verification evidence

Gate	Evidence and what it proves
Python unit suite	93 protocol, compiler, quantized algebra, archive integrity, full-model graph pieces, DMA ordering, and build contracts.
Verilator strict lint	All synthesizable RTL parses with warnings treated as reviewed failures by the project script.
Icarus RTL suite	Four-state simulation of 15 testbenches, including reset, backpressure, boundaries, exhaustive packed MACs, descriptor overrun, and independent fixture comparison.
Full CIFAR-10 software evaluation	95.55% checkpoint and 95.42% exported integer arithmetic over exactly 10,000 official test images.
Vivado OOC implementation	Accelerator-only iteration closes 200 MHz with +0.031 ns setup slack and zero route/DRC errors. Its top-port hold result is explicitly labeled as OOC context because no board part-pin locations exist.
Full PYNQ implementation	Processing system, three clock/reset domains, AXI-Lite and stream CDC, DMA, HP0 DDR path, and accelerator route with +0.005 ns setup and +0.050 ns hold slack; BIT, HWH, XSA, and DCP are hashed.
DRC and CDC audit	Zero DRC errors; 122 named DRC warnings and three AMD interconnect methodology warnings are retained. CDC warnings resolve to AMD XPM asynchronous FIFO internals and their declared timing exceptions.
SAIF power flow	Two XSim activity files are relocated to the exact full-design accelerator hierarchy, matching 9,164 nets before the medium-confidence 1.937 W total estimate.
Physical-board measurement	Supply-rail energy and wall-clock throughput must come from a connected board and are kept distinct from Vivado estimates.

16 Where every line is explained

The companion file `neuraedge_line_by_line_source_reference.tex` contains every RTL, Python, PowerShell, Tcl, and test source file. It includes the complete numbered source and a context-aware annotation for every nonblank line. Its JSON manifest records SHA-256 for every included source, preventing a PDF from being mistaken for documentation of a later code revision.

The 55-page `outputs/deep_debug_final/neuraedge_deep_debug_report.pdf` contains 49 generated evidence plots, all 15 XSim waveform views, artifact hashes, full timing/resource/DRC/CDC/power plots, and the native Vivado `system.bd` export.

The final line-by-line PDF contains 99 source files, 28,915 physical source lines, and an explanation for all 26,565 nonblank lines. Its generated length is 3,277 pages.

17 Primary research and vendor references

- Spikingformer: `https://arxiv.org/abs/2304.11954`
- FlashAttention IO-aware exact attention: `https://arxiv.org/abs/2205.14135`
- I-ViT integer-only vision transformer: `https://arxiv.org/abs/2207.01405`
- FQ-ViT: `https://arxiv.org/abs/2111.13824`
- Alpha-XNOR spiking attention: `https://openaccess.thecvf.com/content/CVPR2025/html/Xiao_Rethinking_Spiking_Self-Attention_Mechanism_Implementing_a-XNOR_Similarity_Calculation_in_Spiking_CVPR_2025_paper.html`
- AMD AXI DMA PG021: `https://docs.amd.com/r/en-US/pg021_axi_dma/Field-Descriptions`
- AMD Vivado power analysis UG907: `https://docs.amd.com/r/en-US/ug907-vivado-power-analysis-optimization/Running-Power-Analysis-from-the-Vivado-IDE`